

CreepPFN: Forecasting Concrete Creep with a Transformer Trained on a Hierarchical Bayesian Prior

Sompote Youwai^{1,*}, Warayut Dokduea² and Weerachart Tangchirapat²

¹AI Research Group, Department of Civil Engineering, King Mongkut's University of Technology Thonburi, Bangkok, Thailand

²Construction Innovations and Future Infrastructures Research (CIFIR), Department of Civil Engineering, Faculty of Engineering, King Mongkut's University of Technology Thonburi, Bangkok, Thailand

*Corresponding author: sompote.you@kmutt.ac.th

Abstract

Laboratory creep tests of concrete often run for months, yet engineers usually need an estimate of long-term creep within days of loading. This study presents CreepPFN, a transformer network that forecasts later creep compliance, with a 90% range, from basic material properties and the first ten days of readings. Measured creep curves are too few to train a model that generalizes to new laboratories, so CreepPFN is first trained on 1.23 million synthetic creep tests drawn from a hierarchical Bayesian model of past tests and then fine-tuned on measured curves. On 610 curves from 65 literature sources of the Northwestern University database, each held out from training, CreepPFN achieved the lowest error of all models tested, with a normalized mean absolute error of 15.94%. It outperformed curve fitting, matched or exceeded tuned machine-learning baselines trained under the same protocol, and was the only model whose 90% ranges kept their nominal coverage on unseen sources. It also gave the lowest error on external laboratory and literature curves. A SHAP analysis showed that strength, modulus and stress ratio adjust the forecast in the directions assumed by the design codes. Because the test sources had been studied before, these results should be regarded as exploratory.

Keywords: Concrete creep; creep test; early-age forecasting; prediction interval; hierarchical Bayesian model; transformer; prior-data fitted network.

Nomenclature

Symbols

t	time since loading (day)
t_a	time of the first reading (day)
$J(t)$	creep compliance ($\mu\epsilon/\text{MPa}$)
$y(t)$	compliance increase after the first reading ($\mu\epsilon/\text{MPa}$)
ρ	concrete density (kg/m^3)
f_c	compressive strength (MPa)
f_{cm}	mean compressive strength (MPa)
E_{28}	28-day elastic modulus (GPa)
E_{rec}	recorded modulus of external curves (GPa)
r_σ	stress ratio, applied stress / strength at loading
k_σ	stress-strength ratio in Eurocode 2
A_k	amplitude of term k of the Kelvin4 curve ($\mu\epsilon/\text{MPa}$)
τ_k	retardation time of Kelvin term k (day)
$\phi_k(t)$	time function of term k of the Kelvin4 curve
$\hat{R}$	Markov chain convergence statistic
R^2	coefficient of determination
r, r_s	Pearson and Spearman correlation

Abbreviations

KMUTT	King Mongkut's University of Technology Thonburi
MAE	mean absolute error
MLP	multilayer perceptron
NMAE	normalized mean absolute error
NU	Northwestern University creep database
PFN	prior-data fitted network
RMSE	root mean square error
SHAP	SHapley Additive exPlanations
XGBoost	extreme gradient boosting

Supplementary Section S1 lists the full notation of the model equations.

1 Introduction

Creep, the slow increase of strain in concrete under sustained load, governs long-term deflection, prestress loss and stress redistribution in concrete structures (Wendner et al., 2015). Its magnitude depends on the concrete mixture, the age at loading, the stress level and the environment, and although design codes estimate creep from these quantities, their predictions scatter widely against measured data (Bažant and Li, 2008). A creep test on the actual concrete removes much of this uncertainty, but the standard compression creep test runs for months or longer (ASTM International, 2015). Engineers, however, often need an estimate within days, for example to accept a mixture, to plan a test programme or to check a design assumption. A method that forecasts the long-term curve from a short test, and states how uncertain that forecast is, would therefore be valuable in practice.

Two approaches exist today, and neither meets this need well. In curve fitting, an engineer fits a creep equation, typically a logarithmic law or a sum of exponential terms, to the early readings and extends it in time. Different equations can match the first ten days equally well yet diverge by a factor of two at day 160, so the chosen equation rather than the data decides the long-term forecast, and the fit provides no uncertainty range. In machine learning, tree ensembles and neural networks have been trained on the Northwestern University (NU) database to predict creep compliance from mixture and test conditions (Liang et al., 2022, Wang et al., 2026). These models do not use the readings of the specimen being tested and need inputs such as loading age, humidity and specimen size that many records lack. Their reported accuracy also often comes from splits in which readings of one publication appear in both training and testing, a practice that Wang et al. showed can inflate accuracy (Wang et al., 2026).

The underlying difficulty is the nature of creep data, which are plentiful in readings but scarce in independent experiments. The 617 usable NU curves come from only 65 literature sources, the largest of which supplies 58 curves while several supply only one, and together they cover a small part of the possible combinations of material, laboratory, specimen and test schedule. A third of them also lack a recorded modulus. A network with 3.1 million weights trained directly on a few hundred such curves would memorize the well-represented sources rather than learn creep behaviour, a known limit of machine learning on small materials datasets (Zhang and Ling, 2018). Its accuracy on familiar sources would also overstate its accuracy on new ones, because curves from one source share unrecorded test conditions (Roberts et al., 2017, Wang et al., 2026). Synthetic training data are therefore a necessity rather than a convenience, provided that they reproduce how real creep curves vary. The results of this study support this view. Machine-learning models trained on measured curves alone reached 90–105% error on new sources without early readings, and even with the readings their calibrated 90% ranges covered only 72–88% of later readings. In contrast, the CreepPFN network trained only on synthetic tests reached 16.12% error before it saw any measured curve.

This study develops such a model, called CreepPFN. The name refers to a prior-data fitted network (PFN), a machine-learning model that learns to forecast by practising on many synthetic tasks (Müller et al., 2022). CreepPFN works in two stages. A hierarchical Bayesian model is first fitted to past creep tests to describe a typical curve for given material properties and how curves scatter between literature sources, between specimens and between readings. A transformer network then practises on synthetic tests drawn from this model, each time forecasting the hidden later readings, with a range, from the early ones. In total, 1,230,000 synthetic tests were used for training, and because the size, shape and scatter of the synthetic curves are estimated from the measured data rather than assumed, these tests follow the distribution of real tests. After this practice, the network is fine-tuned on measured curves and forecasts a new specimen in milliseconds without any refitting, with the Bayesian model supplying general knowledge of creep behaviour and the early readings supplying the specimen’s own level and rate.

Other fields address the same shortage of real data with synthetic data. Networks trained on many randomized simulations transfer to real-world tasks (Tobin et al., 2017), data augmentation reduces overfitting when real examples are scarce (Shorten and Khoshgoftaar, 2019), and networks trained only on synthetic datasets outperform tuned methods on small real tables (Hollmann et al., 2025) and extrapolate real learning curves accurately (Adriaensen et al., 2023). The main risk is that synthetic data may not resemble reality. CreepPFN controls this risk by drawing every synthetic curve from distributions fitted to the measured tests and by calibrating the size of departures from the curve family on validation sources. As a result, the synthetic tests reproduce the variation observed in real tests rather than an assumed one.

The study makes three contributions. First, it presents a forecasting network trained on a creep model estimated from data, rather than on hand-designed synthetic curves or on the measured curves alone. Second, it provides a strict evaluation in which whole literature sources are held out and machine-learning

baselines are retrained and scored under the same protocol, instead of being compared through published numbers. Third, it offers practical value, because the forecast carries a range that keeps its stated coverage on new sources and the accuracy is reported as a function of test duration to support decisions on how long to test.

The paper is organized as follows. Section 2 reviews creep models and data-driven studies and states the research gap, Section 3 describes the CreepPFN framework, and Section 4 describes the data and the evaluation protocol. Sections 5, 6 and 7 present the results, the discussion and the application, and the Supplementary Material gives the equations, model settings and additional results.

2 Related work and research gap

2.1 Engineering creep models

Engineering creep models predict compliance or the creep coefficient before any test is run (Table 1), and the table lists the main inputs of each model and whether measured creep of the same concrete is used to adjust the prediction. Design-code models such as ACI 209R-92, the CEB-FIP Model Code 1990 with Eurocode 2, the *fib* Model Code 2010 and GL2000 write creep as a function of time since loading, with correction factors for strength, age at loading, humidity, member size and cement type (ACI Committee 209, 2008, CEN, 2004, *fib*, 2013, Gardner and Lockman, 2001). Models B3 and B4 rest on the solidification theory of creep and add mixture composition (Bažant and Baweja, 1995, RILEM TC-242-MDC, 2015), and B4 was calibrated on laboratory tests and multi-decade bridge deflections with a quantified uncertainty (Wendner et al., 2015). A statistical comparison on the NU database found clear differences in accuracy between these models and large scatter for all of them (Bažant and Li, 2008), and Bayesian updating has been used to revise model predictions with monitoring data (Sousa et al., 2019). These models remain the engineering reference, but they all require age at loading, humidity and member size, which the present data extract does not record, so this study does not test superiority over them.

Table 1: Engineering creep models and CreepPFN.

Model	Basis	Main inputs	Specimen readings	Output
ACI 209R-92 (ACI Committee 209, 2008)	Empirical code	Age at loading, curing, humidity, volume–surface ratio, slump, cement and air content	Not used	Mean curve
CEB-FIP MC90; Eurocode 2 (CEN, 2004)	Empirical code	Mean strength, age at loading, humidity, notional size, cement type	Not used	Mean curve
<i>fib</i> Model Code 2010 (<i>fib</i> , 2013)	Empirical code	As MC90, plus temperature; basic and drying creep	Not used	Mean curve
GL2000 (Gardner and Lockman, 2001)	Empirical design method	Strength, age at loading and at drying, humidity, volume–surface ratio, cement type	Not used	Mean curve
Model B3 (Bažant and Baweja, 1995)	Solidification theory	Strength, cement content, water–cement and aggregate–cement ratios, humidity, size, age at loading	Optional updating	Mean curve with scatter
Model B4 (RILEM TC-242-MDC, 2015, Wendner et al., 2015)	Extension of B3	As B3, plus admixture, aggregate type and temperature	Bayesian updating studied	Mean curve with uncertainty
CreepPFN (this study)	Hierarchical Bayesian model of past tests, then a trained transformer	Density, strength, 28-day modulus, stress ratio and readings up to day 10	Required	Forecast with 90% range

2.2 Data-driven creep prediction

Data-driven studies have used the NU database to learn creep directly from records, and Table 2 compares their data, inputs, targets and evaluation designs. Because these designs differ, the published scores do not rank the methods, which is why Section 5 retrains comparable model types under one protocol. Liang

et al. trained random forest, XGBoost and LightGBM models on mixture, loading, environment and time and reported R^2 of 0.946–0.953, against 0.377 for the *fib* Model Code 2010 (Liang et al., 2022). Wang et al. trained an FT-Transformer, a transformer for tabular data, on a high-strength-concrete subset with 16 predictors and, splitting the data by database file, reported $R^2 = 0.918$ and a mean absolute error (MAE) of $6.01 \mu\epsilon/\text{MPa}$ (Wang et al., 2026). Other studies combined recurrent networks with attention to extrapolate short tests recursively (Fan et al., 2024), forecast later creep from early readings alone with a convolutional–transformer network called Creepformer (Li et al., 2025), or combined a B4-informed network with Gaussian-process uncertainty (Tao et al., 2026). The authors’ earlier Triple Attention model predicted creep strain from material properties and the reading history of 67 laboratory and literature curves (Dokdua et al., 2025). Together, these studies show that material descriptors, early readings and attention-based networks all carry useful information for creep prediction.

Table 2: Published data-driven creep studies.

Study	Data	Inputs and target	Split and horizon
Liang et al. (Liang et al., 2022)	NU database	Mixture, loading, environment, time; total compliance	Train–test split; interpolation
Wang et al. (Wang et al., 2026)	NU-ITI subset; 7,606 readings, 432 files	16 predictors incl. loading age, humidity, log time; total compliance	Grouped by file; interpolation
Creepformer (Li et al., 2025)	NU database and field data	Early creep readings; later creep	Not stated; extrapolation
Triple Attention (Dokdua et al., 2025)	42 KMUTT and 25 literature curves	Properties and daily history; next-day creep strain	Fitted daily grid; one step ahead
CreepPFN	NU; 610 test curves, 65 sources	Seven descriptors and readings to day 10; compliance increase	Whole sources held out; days 10–160

2.3 Prior-data fitted networks

A PFN is trained on many synthetic tasks drawn from a probability model of plausible data, called the prior, and after training it approximates, in a single fast pass, the forecast that Bayesian inference under that prior would give (Müller et al., 2022). TabPFN and TabICL apply this idea to small tabular datasets (Hollmann et al., 2025, Qu et al., 2025), a learning-curve PFN extrapolates curves from their early part using a prior over curve equations (Adriaensen et al., 2023), and TabPFN-TS and ApolloPFN extend it to time series (Hoo et al., 2026, Potapczynski et al., 2026). All of these priors are designed by hand. For concrete creep, a hand-designed prior would encode the designer’s assumptions about curve shapes and scatter, whereas a prior estimated from past creep tests encodes what the tests actually show.

2.4 Research gap

Four gaps follow from this review. Models with rich material inputs do not use the specimen’s own readings and therefore cannot exploit a short test, while curve fitting uses the readings but no knowledge from other tests, so its long-term shape depends on the chosen equation. Most data-driven studies also report accuracy on random or file-level splits, which do not show how a model performs for a laboratory it has never seen. None of the studies in Table 2 reports a prediction range whose coverage was checked on new sources, although an engineer needs such a range to judge whether a test can stop. All four gaps share one cause, namely that the measured creep curves are too few and too unevenly spread to train a model that generalizes to new laboratories. CreepPFN addresses them together by combining knowledge learned from past tests with the specimen’s early readings, by being evaluated on held-out literature sources against retrained baselines, and by returning a range whose coverage is checked.

3 CreepPFN framework

3.1 Framework overview

Figure 1 shows the flow of CreepPFN, with schematic icons in place of data, while measured examples appear in Section 5. Training (Figure 1A) is carried out once in each evaluation fold, using only that fold’s training sources. It has four steps: a hierarchical Bayesian model is fitted to the measured training curves

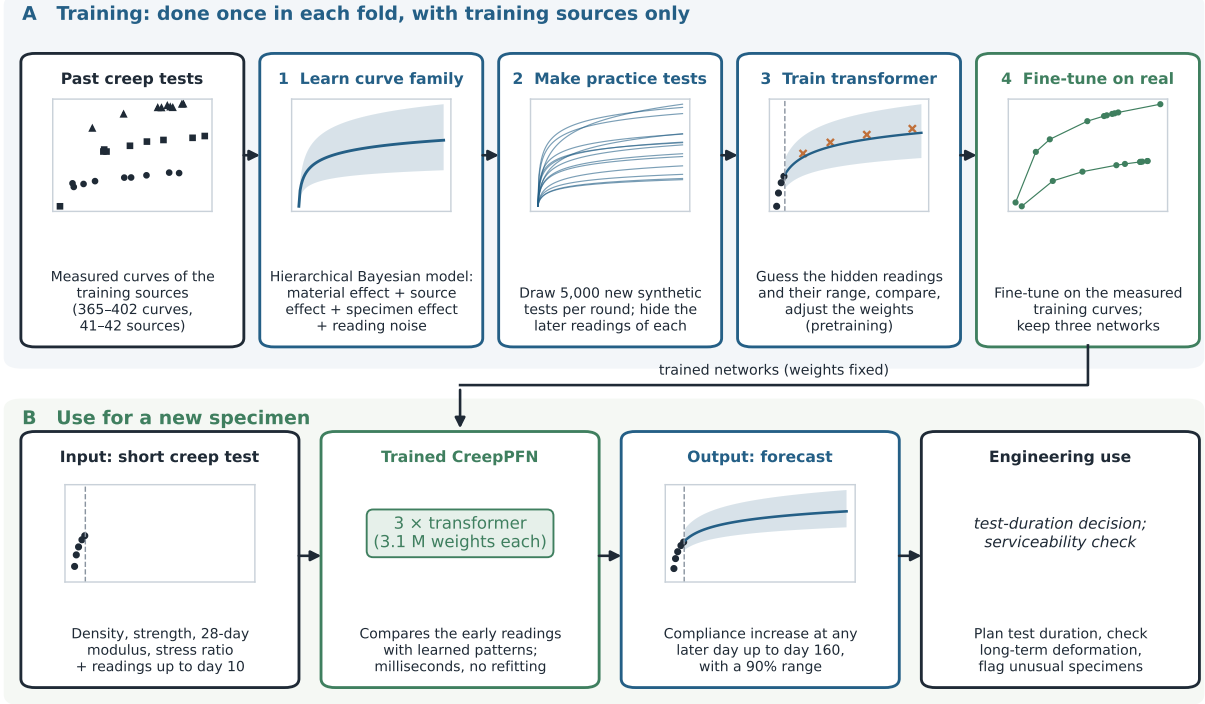


Figure 1: Flow of CreepPFN: (A) training; (B) use for a new specimen.

(Step 1), synthetic practice tests are drawn from it (Step 2), a transformer learns to forecast the hidden later readings of each practice test (Step 3), and the network is fine-tuned on the measured training curves (Step 4). Use (Figure 1B) is simple. The engineer supplies the material record and the readings up to day 10, and the trained network returns the forecast and its 90% range at any requested later day, without fitting an equation or retraining a network for the new specimen.

3.2 Inputs and output

The material inputs are density ρ , compressive strength f_c , the recorded 28-day elastic modulus E_{28} and the stress ratio r_σ , defined as the applied stress divided by the compressive strength at loading, and the creep readings up to day 10 complete the input. Missing modulus or stress-ratio values are flagged and replaced by training medians, and the time of the first reading is also used because some tests start recording after loading. The output is the increase in compliance after the first reading,

$$y(t) = J(t) - J(t_a), \quad (1)$$

where J is the recorded compliance in $\mu\epsilon/\text{MPa}$ and t_a is the time of the first reading in days. Removing the first reading makes curves from different laboratories comparable, because the elastic part and the zero reading differ between test setups. For each requested day, the network returns a central value and a 90% range for y .

3.3 Hierarchical Bayesian model of creep curves

The hierarchical Bayesian model describes how creep curves vary in four layers. The first is a typical curve whose size and shape depend on the seven descriptors through a regression. The second is a source effect that shifts the curves of each literature source to represent unrecorded conditions such as the test rig, the humidity or the mixture. The third is a specimen effect for differences within a source, and the fourth is reading noise. The spread of each layer is estimated from the data rather than set by hand. The model is fitted to all readings of the training curves at once by Hamiltonian Monte Carlo, a standard sampling method for Bayesian models (Hoffman and Gelman, 2014, Phan et al., 2019), so the uncertainty of the fitted parameters carries into the synthetic tests.

Each curve follows a four-term family, called Kelvin4, which writes the compliance increase as

$$y(t) = \sum_{k=1}^4 A_k \frac{\phi_k(t) - \phi_k(t_a)}{\phi_k(160) - \phi_k(t_a)}, \quad \phi_k(t) = 1 - e^{-t/\tau_k} \quad (k = 1, 2, 3), \quad \phi_4(t) = \ln\left(1 + \frac{t}{10}\right), \quad (2)$$

where the retardation times are $\tau_1 = 0.3$, $\tau_2 = 3$ and $\tau_3 = 30$ days and the amplitudes $A_k \geq 0$ add up to the compliance increase at day 160. Each exponential term corresponds to a Kelvin–Voigt element, a spring and dashpot in parallel, and a sum of such terms is the classical Kelvin-chain, or Dirichlet-series, representation of concrete creep (Bažant and Wu, 1973, Bažant and Jirásek, 2018). The three terms thus represent fast early creep, short-term creep and slower creep, while the logarithmic term represents long-term creep that does not level off, as in the logarithmic creep laws of Model B3 and the *fib* Model Code 2010 (Bažant and Baweja, 1995, fib, 2013). Each term is scaled to be zero at the first reading and one at day 160, so that A_k is directly its share of the day-160 increase, and the Bayesian model learns how these amplitudes depend on the material descriptors and scatter between sources and specimens. A common choice adds a 300-day term, but over 160 days that term is almost indistinguishable from the logarithmic term ($r = 0.961$), so the data cannot separate them. With that term the sampler failed to converge (maximum $\hat{R}$ of 1.07–1.82), whereas without it all five folds converged ($\hat{R} \leq 1.065$). A final discrepancy term widens the synthetic curves in proportion to log-time to represent real departures from the Kelvin4 shape that grow with forecast distance, and its size was chosen on validation sources by the quality of the forecasts it implies. Supplementary Sections S2 and S3 give the equations and the diagnostics.

The fitted model also shows why the early readings matter. Across the five folds, the seven descriptors account for only 27–33% of the variance of the log curve size, whereas source effects account for 35–42% (Supplementary Table S5). Two concretes with the same density, strength and modulus can therefore creep very differently: the descriptors set a broad expectation, and the early readings reveal where the specimen actually lies within it.

3.4 Synthetic pretraining and fine-tuning

In Step 2, each practice test copies the material record and reading times of a real training specimen and then draws a source effect, a specimen effect, reading noise and a discrepancy. The resulting curve is evaluated at the recorded times, and its early readings are shown to the network while the later readings are hidden. Tests are drawn in groups of eight that share one source effect, which reproduces the similarity of curves from one publication, and a new set of 5,000 tests is drawn in each training round, so no practice test is repeated. Pretraining ran for 10 to 25 rounds per network. Each network therefore saw 50,000 to 125,000 synthetic tests, with a median of 85,000 and the kept checkpoint reached after a median of 55,000, and the 15 networks of the five folds used 1,230,000 synthetic tests in total (Supplementary Table S7). These tests add variety rather than new experiments, because their material records and reading times come from the measured training curves and their scatter follows the spreads estimated by the Bayesian model. They therefore stay within the range of materials and schedules seen in training.

In Step 3, the network learns from these practice tests. It is a transformer, the architecture behind modern language models (Vaswani et al., 2017), with 3.1 million weights, and Figure 2 shows it in simple blocks. Each early reading becomes a numerical token to which a summary of the material record is added. A context encoder then relates the readings to one another through attention, which weighs each reading in the context of the others so that an unusual reading does not dominate the forecast. For each requested later day, cross-attention takes the relevant information from the encoded readings, and an output head returns a central value and a spread. The network is scored by how well this value and range match the hidden readings, and its weights are adjusted to improve the score. Pretraining took about one minute per network on one GPU, and Supplementary Section S4 and Eq. S9 give every component and the equations.

In Step 4, the network is fine-tuned on the measured training curves with a small learning rate. The pretrained network already reached 16.12% on the test sources before this step, and fine-tuning changed the error by only -0.18 points, so most of the forecasting skill comes from the practice tests. Three networks are trained with different random seeds, the final forecast averages their central values, and the 90% range covers the middle 90% of their combined distribution (Supplementary Sections S4 and S5).

3.5 Differences from existing approaches

Table 3 contrasts CreepPFN with the three existing approaches. Design codes and tabular machine learning predict creep before testing without using the specimen’s readings, whereas curve fitting uses the readings

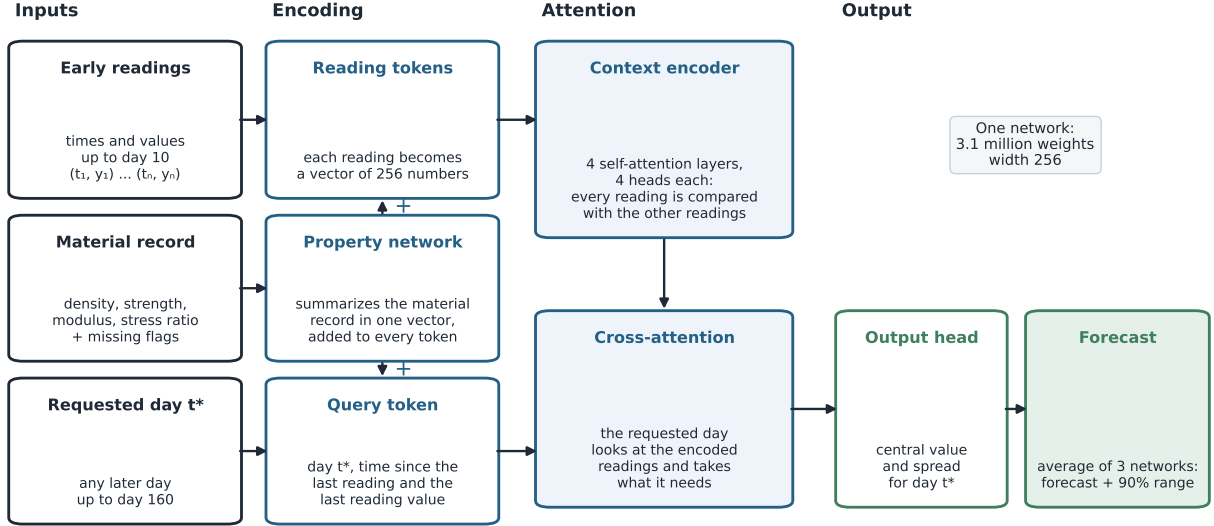


Figure 2: Architecture of one CreepPFN network.

but no knowledge from other tests, so its long-term shape depends on the chosen equation. CreepPFN uses both sources of information: the Bayesian model supplies knowledge of how curves vary, the early readings place the specimen within that variation, and the network returns a range learned from the scatter of past tests without any fitting for the new specimen. Unlike previous PFNs, its prior is estimated from measured tests rather than designed by hand, and unlike the earlier Triple Attention model, which predicted the next value on a smoothed curve, it forecasts measured later readings from a fixed short test.

Table 3: CreepPFN compared with existing approaches.

Feature	Design codes	Tabular machine learning	Curve fitting of early readings	CreepPFN
Uses the specimen's early readings	No	No	Yes	Yes
Uses knowledge from other tests	Calibrated formulas	Learned from records	No	Learned from a Bayesian model of past tests
Required material inputs	Many, incl. age at loading, humidity, size	Many, incl. mixture and conditions	None	Density, strength, modulus, stress ratio
Long-term shape set by	Code formula	Training records	Chosen equation	Learned variation of past curves
Uncertainty range	Mean curve or model scatter	Usually none	None	90% range, coverage checked on new sources
Work for a new specimen	Evaluate formula	Run model	Fit equation	Run model (milliseconds)

4 Data and evaluation protocol

4.1 Creep data

The main data set is a curated extract of the NU creep database (Hubler et al., 2015) containing 617 curves from 65 literature sources and 9,790 readings up to day 160. Density ranges from 1,106 to 2,632 kg/m³ and strength from 10.8 to 167 MPa, the modulus is missing for 34.5% of curves, and the stress ratio is missing for 184 of the 610 evaluated curves. A typical curve has five readings up to day 10 and seven later readings at irregular times. The literature source is the grouping unit, although different publications do not necessarily represent different laboratories, and the extract does not record humidity, age at loading, specimen size or mixture proportions. Two external sets provide an additional test: 47 curves from the Concrete and Materials Testing Laboratory of King Mongkut's University of Technology Thonburi (KMUTT), tested to ASTM C512 (ASTM International, 2015), and 19 curves compiled from published

studies (Rossi et al., 2013, Choi et al., 2025, Hong et al., 2024, Zhao et al., 2016, Gao et al., 2025). The origin of each external curve was checked against the laboratory’s provenance workbook, and six curves that an earlier compilation had placed in the literature set were counted as KMUTT curves because their properties and readings match laboratory mixes exactly. External strains were converted to compliance with a nominal stress of $0.4f_c$, and their stress ratio is treated as missing (Supplementary Section S6).

4.2 Evaluation protocol

The evaluation holds out whole literature sources. The 65 sources were split into five folds, and in each fold the test sources were withheld, about 20% of the remaining sources formed a validation set, and the rest formed the training set. The Bayesian model, the network and all baselines were fitted on the training sources only, settings were chosen on the validation sources, and the test sources were scored once. Every curve with at least two readings up to day 10 and one later reading was tested, giving 610 curves and 5,646 later readings. All methods performed the same task of forecasting all later readings up to day 160 from the readings up to day 10.

4.3 Baselines

Three groups of baselines were used. The first fits the early readings only, through persistence, which holds the last reading constant, a logarithmic law and a Kelvin-plus-log sum with nonnegative coefficients. The second represents earlier NU studies and consists of XGBoost (Chen and Guestrin, 2016), a random forest and a multilayer perceptron (MLP) trained on the seven descriptors and time without specimen readings. The third gives the same three learners the day-10 readings as extra inputs, including the last reading, its time, the number of readings and the forecasts of the logarithmic and Kelvin fits. This group is the strongest fair competitor because it sees exactly the information that CreepPFN sees. Each learner was tuned on the validation sources, its 90% range was calibrated on the validation curves, and a quantile version of XGBoost provided a second, direct range. Finally, the same network trained on an earlier, simpler synthetic generator (Dokdua et al., 2025) shows what the Bayesian model adds (Supplementary Section S9).

4.4 Error and coverage

Error is measured by the normalized mean absolute error (NMAE). For each curve, the mean absolute forecast error over its later readings is divided by the mean absolute value of those readings. Curve errors are then averaged within each source before sources are averaged with equal weight, so that no large publication dominates the score. Coverage, the share of later readings inside the 90% range, is averaged in the same way and should be close to 90% for a well-calibrated range. Confidence intervals and paired differences between methods come from 2,000 bootstrap resamples of the 65 test sources, and pooled R^2 , root mean square error (RMSE) and MAE are also reported for comparison with earlier studies (Supplementary Section S2).

4.5 Input attribution by SHAP

SHAP values (SHapley Additive exPlanations) measure how much each input moves a forecast away from a reference forecast (Lundberg and Lee, 2017). Here they show how the material descriptors adjust the forecast of a specimen whose early readings are known, so the readings and the first-reading time were held at the specimen’s own values. The four descriptor groups were density, strength, the modulus with its missing-value flag and the stress ratio with its missing-value flag. For each test curve, the fold model that withheld it evaluated all 16 combinations of these groups, with replaced groups taken from background curves drawn one per training source. The forecasts at days 28, 90 and 160 were explained, so no later measurements were used, and a second, independently drawn background checked the sensitivity to this choice. SHAP values describe how the model uses its inputs rather than causal material effects, and correlated inputs such as strength and modulus share credit (Supplementary Section S8).

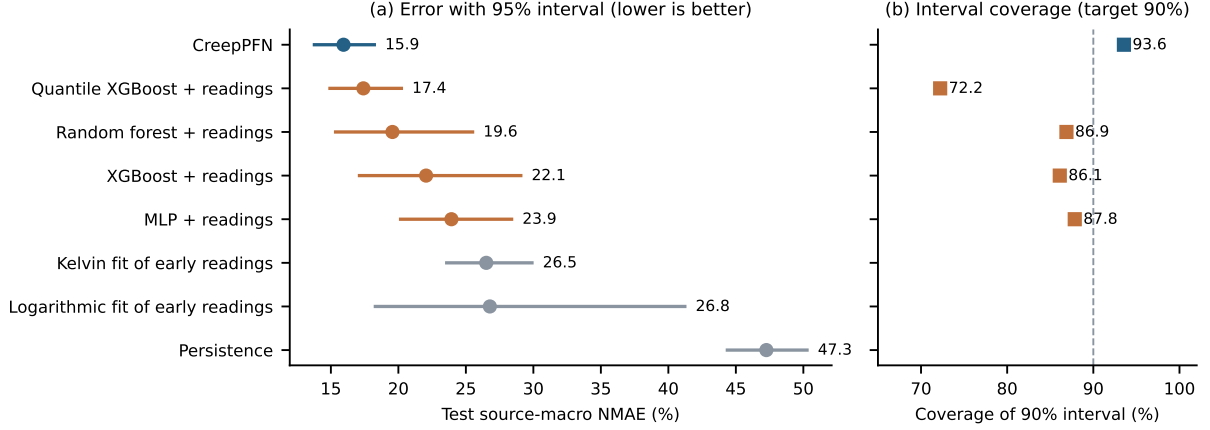


Figure 3: Test error (a) and coverage of the 90% range (b) on 65 held-out sources.

5 Results

5.1 Forecast accuracy

CreepPFN had the lowest error of all models (Figure 3, Table 4), with an NMAE of 15.94% (95% interval 13.78–18.22%). In Figure 3, blue marks CreepPFN, orange the machine-learning models given the same day-10 readings and grey the curve fits, which give no range, while Table 4 adds the pooled R^2 , RMSE and MAE in $\mu\epsilon/\text{MPa}$ and the paired difference between CreepPFN and each model. Among the learners given the day-10 readings, quantile XGBoost reached 17.41%, the random forest 19.56%, XGBoost 22.05% and the MLP 23.93%, and the curve fits reached 26.51% for the Kelvin fit, 26.78% for the logarithmic law and 47.26% for persistence. CreepPFN was significantly more accurate than XGBoost (difference -6.12 points, interval -13.30 to -1.65), the MLP (-7.99 points, interval -12.18 to -4.48) and the Kelvin fit (-10.57 points, interval -13.58 to -7.67). Against quantile XGBoost and the random forest, the differences of -1.47 and -3.62 points had intervals that included zero (-3.65 to +0.28 and -9.46 to +0.34), so CreepPFN is more accurate than these two models on average but not significantly so. Each baseline was trained with one seed, whereas CreepPFN averages three networks. Pooled over readings, CreepPFN reached $R^2 = 0.908$, RMSE $10.50 \mu\epsilon/\text{MPa}$ and MAE $5.22 \mu\epsilon/\text{MPa}$, the last being $0.01 \mu\epsilon/\text{MPa}$ higher than that of quantile XGBoost ($5.21 \mu\epsilon/\text{MPa}$). Averaged by source, as for NMAE, the MAE of CreepPFN was lower by $-0.47 \mu\epsilon/\text{MPa}$ (interval -1.82 to $+0.44$), so the two models have equal absolute error within sampling uncertainty.

Table 4: Same-protocol comparison on 610 test curves from 65 sources.

Model	NMAE (%)	Cov. (%)	R^2	RMSE	MAE	Difference (pp)
CreepPFN (proposed)	15.94 [13.78, 18.22]	93.6	0.91	10.50	5.22	—
Quantile XGBoost + day-10 readings	17.41 [14.93, 20.22]	72.2	0.85	13.39	5.21	-1.47 [-3.65, +0.28]
Random forest + day-10 readings	19.56 [15.35, 25.50]	86.9	0.83	14.17	5.55	-3.62 [-9.46, +0.34]
XGBoost + day-10 readings	22.05 [17.12, 29.07]	86.1	0.82	14.71	5.73	-6.12 [-13.30, -1.65]
MLP + day-10 readings	23.93 [20.16, 28.39]	87.8	0.78	16.41	7.52	-7.99 [-12.18, -4.48]
Random forest, descriptors + time	90.11 [62.61, 132.15]	88.4	0.15	31.96	18.57	-74.18 [-118.86, -46.37]
XGBoost, descriptors + time	94.08 [66.84, 131.98]	87.4	0.19	31.16	19.36	-78.15 [-118.45, -51.34]
Quantile XGBoost, descriptors + time	92.84 [67.01, 127.97]	50.1	0.17	31.58	19.68	-76.91 [-114.11, -52.19]
MLP, descriptors + time	105.47 [72.99, 149.72]	89.3	-0.28	39.16	24.25	-89.53 [-134.46, -57.50]

Note: Bold marks the best value in each column: the lowest NMAE, RMSE and MAE, the highest R^2 , and coverage at or above the nominal 90%. An interval in the last column that excludes zero indicates a significant difference. The pooled MAE of quantile XGBoost is lower by only $0.01 \mu\epsilon/\text{MPa}$; averaged by source, CreepPFN has the lower MAE (difference $-0.47 \mu\epsilon/\text{MPa}$, not significant). NMAE = normalized mean absolute error; Cov. = coverage of the 90% range; RMSE = root mean square error; MAE = mean absolute error; MLP = multilayer perceptron.

The models without specimen readings failed on new sources, reaching 90–105% NMAE and $R^2 \leq 0.19$ with descriptors and time only. This result is consistent with the Bayesian model, in which the descriptors explain only about a third of the variation in curve size, and it shows that a short test carries information that material properties alone cannot supply.

5.2 Calibration of the 90% range

CreepPFN was the only model whose range kept its nominal coverage on new sources, covering 93.6% of test readings with a mean width of $30.4 \mu\epsilon/\text{MPa}$. The calibrated ranges of the baselines covered 86.1–87.8% despite being calibrated on validation sources, and the direct quantile ranges of XGBoost covered only 72.2%. The coverage of CreepPFN also stayed close to nominal at every forecast distance, from 96.9% for readings within 14 days of the last input reading to 91.3% for readings 90–160 days beyond it (Supplementary Table S11). Meanwhile, the range widened with distance, as an honest extrapolation should. For an engineer, the width of the range can therefore be read as a realistic statement of uncertainty.

5.3 Individual forecasts

Figure 4 shows nine forecasts from readings up to day 10, marked by the dashed line. Black points are the input readings, orange crosses are later readings used only for scoring, the line is the forecast and the shading is the 90% range, while each panel gives the density (kg/m^3), strength (MPa), modulus, stress ratio and NMAE of its curve. The NU panels are the first five of six curves chosen from evenly spaced source identifiers. The sixth, NU00826, had only three readings up to day 10 and an NMAE of 29.1%, at the 85th percentile of test curves, and it is omitted from the figure, with its error reported here instead. The KMUTT panels show the first and last KMUTT curves, and the literature panels show the literature curve with the median error, PA048, and the last literature curve, PA065. PA065 probably comes from Zhao et al. (Zhao et al., 2016), whose reported 28-day modulus its block matches, and PA048 most likely comes from the recycled-aggregate-concrete compilation of Gao et al. (Gao et al., 2025), as Section 7.3 explains. Each NU curve is forecast by the fold model that withheld its source, and the external curves by the fold-1 model. Panel errors range from 5.4% to 31.5% around the median test-curve error of 13.1%, and the largest errors occur where a curve changes its trend after day 10. NU00599 and NU00839 flatten more than forecast, whereas literature curves that keep rising strongly give the opposite error. The ten-day readings give little sign of such changes, and every baseline misses them too, but in these cases most measured readings still fall inside the forecast range, which is the behaviour an engineer needs from an uncertain forecast.

5.4 Effect of test duration

Longer tests gave steadily better forecasts (Figure 5). This analysis used the 556 test curves with readings at every duration, all scored on the same readings after day 28. The NMAE of CreepPFN fell from 25.40% after 3 days to 18.64% after 7, 17.20% after 10, 14.80% after 14 and 9.58% after 28 days, so extending a test from 10 to 28 days reduced the error by 44%. Coverage stayed between 93% and 96% at every duration (Figure 5b). CreepPFN was also more accurate than the logarithmic fit at every duration, most clearly for short tests (25.40% against 43.88% after 3 days), when a curve fit has only a few readings to work with.

5.5 Contribution of the material descriptors

Strength had the largest effect on the forecast among the descriptors (Figure 6). Figure 6a shows the mean absolute contribution of each descriptor at days 28, 90 and 160, averaged within and then over sources, with black ticks for the second background panel. The contributions are in $\mu\epsilon/\text{MPa}$ and describe model behaviour rather than causal effects. At day 90, strength accounted for 44.4% of the total absolute descriptor contribution, with a mean of $2.30 \mu\epsilon/\text{MPa}$, followed by the modulus group (31.2%), the stress-ratio group (21.8%) and density (2.6%). Together, the descriptors moved the day-90 forecast by $5.17 \mu\epsilon/\text{MPa}$ on average, about 11% of the mean forecast of $46.8 \mu\epsilon/\text{MPa}$. Their effect grew from $2.94 \mu\epsilon/\text{MPa}$ at day 28 to $6.37 \mu\epsilon/\text{MPa}$ at day 160 because the early readings constrain the far end of the curve less. The second background panel changed no share by more than 0.1 percentage points.

Figure 6b shows the direction of each contribution for individual curves. For strength and modulus, red points (high values) lie mostly left of zero and blue points (low values) to the right, whereas the stress ratio shows the reverse pattern and grey points (missing values) cluster just left of zero. These directions agree with creep behaviour. Higher strength lowered the forecast compliance, with a Spearman correlation r_s of -0.94 between strength and its SHAP value over 610 curves, and so did a higher recorded modulus (r_s of -0.73 , 398 curves). A higher stress ratio raised it (r_s of $+0.63$, 426 curves), consistent with larger creep per unit stress at higher load levels, and higher density lowered it slightly (r_s of -0.21), although

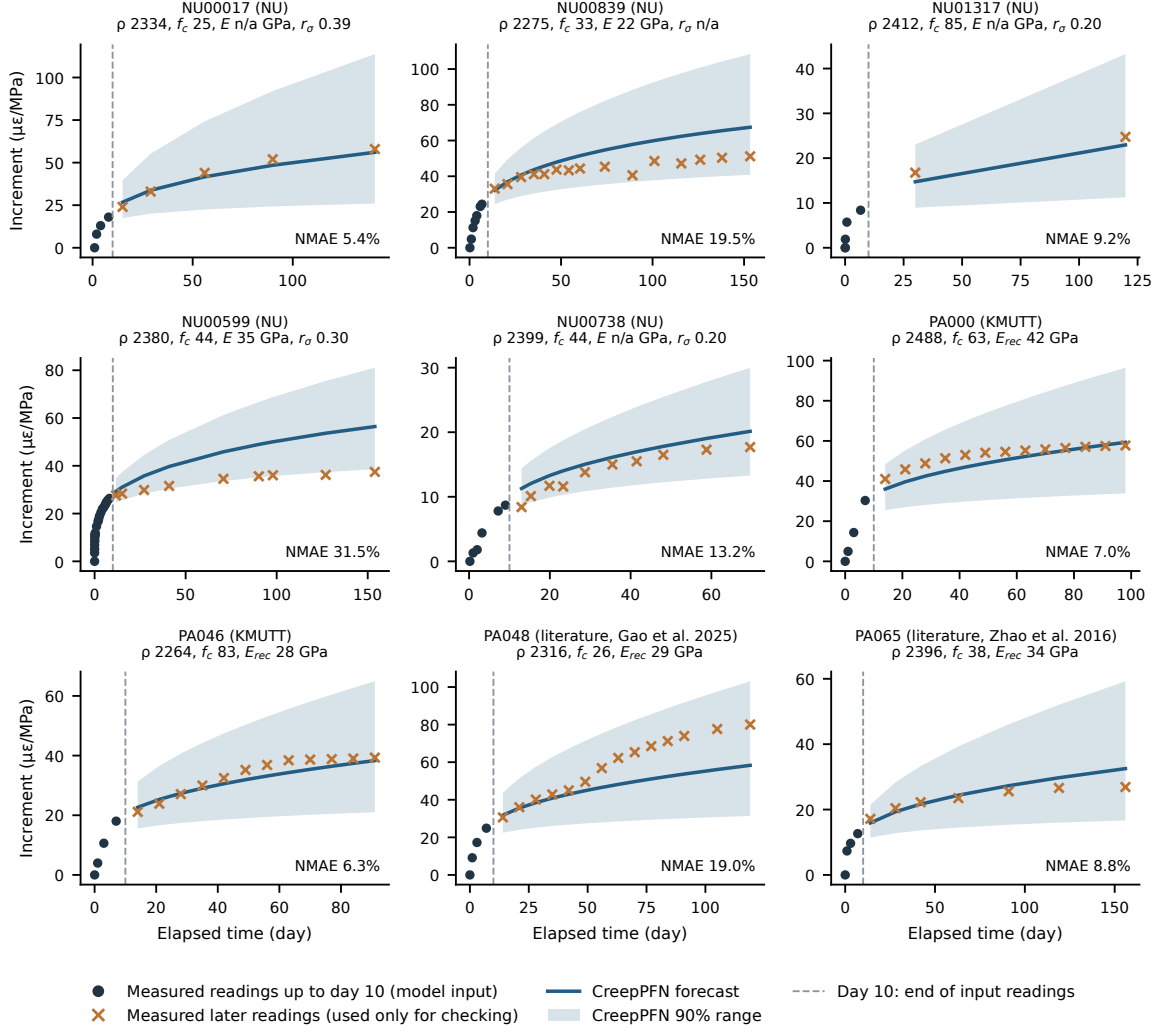


Figure 4: Nine forecasts from readings up to day 10.

density had the smallest effect. A missing stress ratio changed the day-90 forecast by $-1.39 \mu\epsilon/\text{MPa}$ on average, against $+1.37 \mu\epsilon/\text{MPa}$ when the ratio was recorded. The model has therefore learned that records without a stress ratio tend to show less creep, which is a pattern of the sources in the database rather than a material effect.

5.6 Contribution of the hierarchical Bayesian model

The same network trained on an earlier, simpler synthetic generator, which fitted curve equations to each training curve, regressed their parameters on the descriptors and resampled the residuals, reached 17.12% NMAE. The Bayesian model lowered the error by -1.18 points, but the interval of -2.40 to $++0.03$ just includes zero, so the gain in accuracy is suggestive rather than proven. Its clearer benefits are that it separates source, specimen and reading variation, uses an identifiable curve family and lets the pretrained network reach 16.12% before it sees any measured curve (Supplementary Table S10).

6 Discussion

6.1 Sources of forecasting performance

CreepPFN performs well because it combines two kinds of evidence that other approaches use separately. The Bayesian model provides a realistic spread of curve shapes and sizes for a given material, including how much laboratories differ, and the early readings then fix the specimen’s own level and early rate.

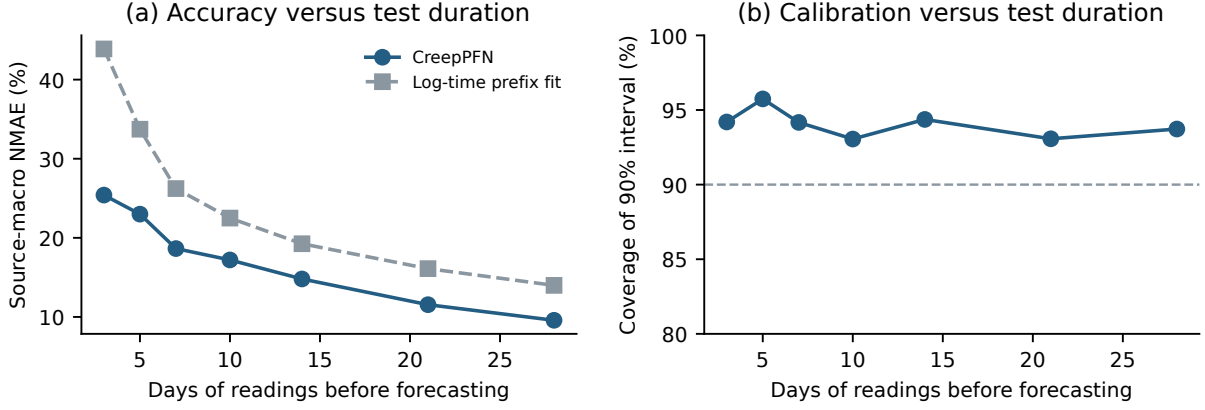


Figure 5: Forecast error (a) and coverage (b) against test duration.

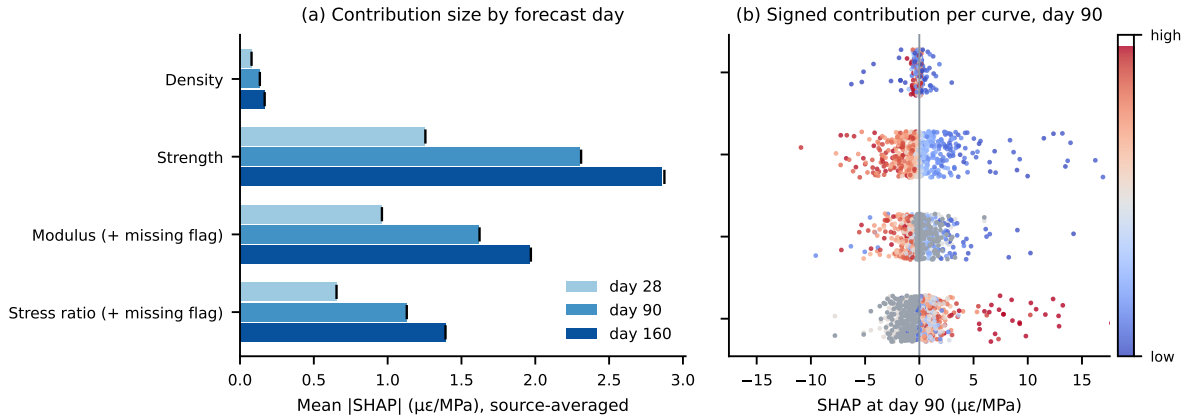


Figure 6: Contributions of the material descriptors (SHAP).

A curve fit has only the second kind of evidence, so its long-term shape rests on an assumed equation, and a tabular model has only the first, so it cannot tell two similar concretes apart. The SHAP analysis confirms that, once the readings are known, the descriptors adjust the day-90 forecast by about 11% in physically sensible directions. The machine-learning models given the early readings have both kinds of evidence, which explains why quantile XGBoost and the random forest came close in accuracy. They learned from only 365–402 measured curves per fold, however, whereas CreepPFN learned from tens of thousands of practice tests that reproduce the scatter between sources. It is this scatter that matters for a new laboratory and explains the better calibration of CreepPFN’s ranges.

The calibration result is the most useful one for practice. The 90% range of quantile XGBoost covered only 72% of readings, which would lead an engineer to stop tests too early or to trust deflection estimates too much. The ranges of CreepPFN, in contrast, held 91–97% coverage across horizons and test durations on sources the model had never seen. This reliability comes from the training, because the practice tests contain source effects and forecast discrepancy, so the network learns how wrong an extrapolation can be for a new source.

6.2 Interpretation of the descriptor effects

The descriptor effects in Figure 6 follow the dependencies that the design codes build into their creep models. Eurocode 2 gives the creep strain as the creep coefficient multiplied by the stress and divided by the tangent modulus (CEN, 2004), so creep per unit stress falls as the modulus rises, and it also reduces the notional creep coefficient with the strength factor $16.8/\sqrt{f_{cm}}$ (CEN, 2004). In the *fib* Model Code 2010, the basic and drying creep coefficients decrease with the mean strength as $1.8/f_{cm}^{0.7}$ and $412/f_{cm}^{1.4}$, respectively (fib, 2013). Both codes therefore predict less creep per unit stress for stronger and stiffer concrete. CreepPFN learned the same two directions from data without being given any of these formulas,

with Spearman correlations of -0.94 and -0.73 between the strength and modulus contributions and the recorded values. Models B3 and B4 express the same trend through the water–cement ratio and the cement content, which are closely linked to strength (Bažant and Baweja, 1995, RILEM TC-242-MDC, 2015). Because strength and modulus are correlated in the data, their SHAP shares should be read together, and their combined share was about 76% of the descriptor effect at day 90.

The stress-ratio effect agrees with the known nonlinearity of creep at high load levels. Creep is close to proportional to stress at low stress ratios. Eurocode 2, however, requires non-linear creep to be considered when the stress exceeds 0.45 of the characteristic strength at loading, and it then multiplies the creep coefficient by $\exp[1.5(k_\sigma - 0.45)]$, where k_σ is the stress–strength ratio (CEN, 2004). The positive stress-ratio contributions of CreepPFN (r_s of $+0.63$) are consistent with this behaviour, although the model does not separate the linear and nonlinear ranges. The small density effect is also plausible, because Eurocode 2 scales both the modulus and the creep coefficient of lightweight concrete by the same density factor (CEN, 2004), so the creep compliance, which is their ratio, depends little on density. In CreepPFN, density accounted for only 2.6% of the descriptor effect.

These results agree with earlier SHAP studies on the NU database, with one important difference. Liang et al. found time since loading, compressive strength, age at loading, relative humidity and temperature most influential in their tree ensembles (Liang et al., 2022). Wang et al. identified loading age, loading duration, the aggregate–cement ratio and humidity (Wang et al., 2026), and Bouras and Li found time to dominate, followed by loading age, strength, modulus, volume–surface ratio and humidity (Bouras and Li, 2023). Strength and modulus therefore rank among the leading material inputs in every study, as they do here. In those studies, however, time and test conditions dominated because the models predict the whole curve from inputs alone. In CreepPFN, by contrast, the early readings already fix the level and early rate of the curve and the descriptors only adjust the forecast, by about 11% at day 90. Age at loading, humidity and specimen size, which ranked highly in the earlier studies, are not recorded in the present data, so their effects are absorbed by the early readings and the source effects and cannot be assessed by this SHAP analysis.

6.3 Comparison with previous studies

The published NU studies report higher R^2 than CreepPFN, but they answer a different question. Liang et al. and Wang et al. predict total compliance from mixture and test conditions (Liang et al., 2022, Wang et al., 2026), and total compliance includes the elastic part, which varies strongly between concretes and is easy to explain, so it raises R^2 . CreepPFN predicts the increase after the first reading, which removes that easy part. Those models also use loading age, humidity and size, which the present extract lacks, and most of them interpolate within the time range of each training curve. The closest comparison is the FT-Transformer of Wang et al., which also held out whole files: its MAE of $6.01 \mu\epsilon/\text{MPa}$ is higher than the $5.22 \mu\epsilon/\text{MPa}$ of CreepPFN, and its RMSE of $9.06 \mu\epsilon/\text{MPa}$ is lower than the $10.50 \mu\epsilon/\text{MPa}$ of CreepPFN. Because the data subsets and targets differ, this comparison indicates similar accuracy rather than a ranking, which is why Section 5 retrained the same model types under one protocol.

Pooled R^2 is a weak measure for creep curves. The Kelvin fit of the early readings, which learns nothing from other tests, reached $R^2 = 0.868$ on the test readings, yet its NMAE was 26.51% against 15.94% for CreepPFN, because smooth increasing curves give a high R^2 to almost any reasonable extrapolation. Source-averaged error, range coverage and paired comparisons with baselines are therefore more informative. Similarly, the 1.63% error reported for the Triple Attention model (Dokdua et al., 2025) refers to next-day prediction on a curve smoothed by a fitted equation, where simply repeating the previous value scores about 1.7%, a far easier task than a 150-day forecast of measured readings.

6.4 Limitations

Several limitations qualify these results. The test sources had been examined in earlier studies of these data, so the results are a reassessment rather than an independent confirmation, and the baselines were trained with one seed, whereas CreepPFN averages three networks. Two of the five Bayesian fits reached a maximum $\hat{R}$ slightly above 1.05, and no model trained on measured curves alone with the same network was evaluated, so the benefit of practice tests over direct training is not isolated. The data extract lacks humidity, age at loading, specimen size and mixture composition, and the Bayesian model represents increasing, concave curves without unloading, varying environments or tertiary creep. Finally, the external set contains one laboratory and one partly unresolved literature set with an assumed stress, and the 90%

range applies at each day separately rather than as a band for the whole curve.

7 Application

7.1 Laboratory implementation

Applying CreepPFN requires four steps. The engineer first records density, compressive strength, the 28-day modulus and the stress ratio, marking any missing value. The engineer then supplies at least two creep readings up to day 10, with times in days since loading and strain converted to compliance with the applied stress. After the days of interest up to day 160 are requested, the model returns the forecast and the 90% range at those days. This takes about 18 ms for one curve on a laptop processor or about 4 ms per curve when many curves are processed together. The output is the creep compliance increase of the tested specimen, so a structural deformation estimate still requires the stress history, the geometry, shrinkage and the elastic strain.

7.2 Test-duration planning

The duration results support a simple planning rule, namely to choose the shortest test whose expected error meets the project tolerance. On the NU test sources, a 7-day test gave about 18.64% error, a 14-day test about 14.80% and a 28-day test about 9.58%. The range can also be used after each reading: the test can stop once the forecast range at the design age is narrow enough for the decision at hand, and otherwise it continues while the forecast is updated with each new reading. These tolerances are indicative, because they come from literature sources, and they should be confirmed for a given laboratory before routine use.

7.3 External validation

The frozen fold models were applied to the 66 external curves without any adjustment (Table 5). The table gives the curve-averaged NMAE, averaged over the five fold models, with the range across folds in brackets, and the curve fits are the same in every fold. The stress ratio is missing for every external curve, and the literature responses assume a stress of $0.4f_c$. On the 47 KMUTT curves, CreepPFN reached 11.08% NMAE with 99.2% coverage, and on the 19 literature curves it reached 21.90% with 74.2% coverage, against 18.82% and 23.54% for the logarithmic fit. The earlier CreepPFN, trained on the empirical generator, reached 10.15% on the KMUTT curves and 24.27% on the literature curves. Paired over curves, the proposed model was significantly more accurate than the earlier network on the literature curves, by -2.37 points (interval -4.40 to -0.66). The difference of $+0.94$ points on the KMUTT curves (interval -0.11 to $+1.97$) was not significant, and the pooled difference was -0.01 points (interval -1.04 to $+0.94$). The hierarchical Bayesian prior therefore helped most on the literature curves, which differ most from the NU training sources, and both networks clearly outperformed the curve fits.

The literature error comes mainly from one block of curves. The ten curves PA047–PA056 averaged 31.6% NMAE, compared with 11.0% and 11.2% for the curves that probably come from Hong et al. (Hong et al., 2024) and Zhao et al. (Zhao et al., 2016). All ten keep rising strongly after day 20, with a late creep rate about half the early rate, against about a third for the KMUTT curves. By elimination among the cited sources, these curves most likely come from the recycled-aggregate-concrete compilation of Gao et al. (Gao et al., 2025). Recycled-aggregate concrete creeps more than normal concrete and keeps creeping longer, which is consistent with the strong late creep of these curves, and the NU training data contain almost no such concrete. An unknown loading stress, which the assumed $0.4f_c$ conversion cannot represent, may add to the error. The model under-predicts all ten curves, which also explains the literature coverage below nominal, and it has not been shown to handle such curves.

Table 5: External forecasts from readings up to day 10.

Method	KMUTT (47)	Other literature (19)	Pooled (66)
CreepPFN (proposed)	11.08 [10.06, 12.00]	21.90 [20.01, 22.88]	14.20 [12.92, 14.90]
Earlier CreepPFN (empirical generator)	10.15 [8.76, 11.83]	24.27 [23.26, 25.93]	14.21 [13.26, 15.75]
Logarithmic fit of early readings	18.82	23.54	20.18
Kelvin fit of early readings	26.56	32.55	28.29
Persistence	45.26	56.37	48.45

Note: Bold marks the lowest NMAE in each column. Brackets give the range over the five fold models. The proposed model is significantly more accurate than the earlier network on the literature curves; the other differences between the two networks are not significant (paired bootstrap over curves; see text). Both networks are significantly more accurate than the curve fits.

8 Conclusions

This study presented CreepPFN, a transformer network that forecasts the creep compliance of concrete, together with a 90% range, from density, strength, modulus, stress ratio and the readings of the first ten days of a creep test. Because the available creep curves are too few and too unevenly spread to train a model that generalizes to new laboratories, the network was first trained on 1,230,000 synthetic creep tests drawn from a hierarchical Bayesian model of past tests. This model, built on an identifiable Kelvin-chain curve family, separates the variation between materials, literature sources, specimens and readings and estimates each part from measured data. The network was then fine-tuned on measured curves, and it forecasts a new specimen in milliseconds without any refitting.

Evaluated on 610 curves from 65 literature sources that were withheld from training, CreepPFN had the lowest error of all models tested, with a source-averaged NMAE of 15.94%. It was significantly more accurate than gradient boosting, a neural network and curve fitting of the early readings, and it matched quantile gradient boosting and a random forest that were trained and tuned under the same protocol. Its clearest advantage lies in the reliability of its uncertainty, since it was the only model whose 90% range kept its nominal coverage on unseen sources, at 93.6% against 72.2–87.8% for the baselines, and its coverage stayed close to nominal across forecast distances and test durations. Models that used material properties without the specimen’s readings failed on new sources, which confirms that a short test carries information that material data alone cannot provide.

The results also have direct practical and physical meaning. Forecast error fell from 25.40% with three days of readings to 17.20% with ten days and 9.58% with 28 days, which gives laboratories a quantitative basis for choosing how long to test. On external curves, CreepPFN reached 11.08% NMAE for the KMUTT laboratory and was significantly more accurate than the earlier network on the literature curves, although curves that probably come from recycled-aggregate concrete remained poorly forecast. The SHAP analysis showed that higher strength and modulus lower the forecast creep while a higher stress ratio raises it, in agreement with the dependencies built into Eurocode 2 and the *fib* Model Code 2010, even though the model was never given these formulas.

These findings remain exploratory, because the test sources had been examined in earlier studies, the external metadata are incomplete, and the data extract lacks humidity, loading age and specimen size. Future work should therefore test CreepPFN on a newly reserved set of creep tests with recorded loading age, applied stress and environment, extend the prior to recycled-aggregate and other concretes that are rare in the training data, and isolate the benefit of synthetic pretraining against direct training on measured curves. With such confirmation, CreepPFN could allow laboratories to shorten creep tests and to report long-term creep together with an uncertainty range that engineers can trust.

Data and code availability

The code, fitted models and outputs are available in the project repository. The Supplementary Material gives the equations, the Bayesian model settings and diagnostics, the network architecture, the training budgets, the data preparation, the baseline settings and additional results.

Supplementary Material

Supplementary Material accompanies this article. It contains the notation, the equations (Eqs. S1–S11), the settings and diagnostics of the Bayesian model, the network architecture, the training budgets, the data details, additional results and the baseline settings.

References

- ACI Committee 209 (2008). *Guide for Modeling and Calculating Shrinkage and Creep in Hardened Concrete* (ACI 209.2R-08). American Concrete Institute, Farmington Hills, MI. <https://www.concrete.org/publications/internationalconcreteabstractsportal/m/details/id/19835>.
- Adriaensen, S., Rakotoarison, H., Müller, S., and Hutter, F. (2023). Efficient Bayesian learning curve extrapolation using prior-data fitted networks. *Advances in Neural Information Processing Systems*, 36. arXiv:2310.20447. <https://doi.org/10.48550/arXiv.2310.20447>.
- ASTM International (2015). Standard test method for creep of concrete in compression. ASTM C512/C512M-15. https://doi.org/10.1520/C0512_C0512M-15.
- Bazant, Z.P., and Baweja, S. (1995). Creep and shrinkage prediction model for analysis and design of concrete structures—model B3. *Materials and Structures*, 28, 357–365. <https://doi.org/10.1007/BF02473152>.
- Bazant, Z.P., and Li, G.-H. (2008). Unbiased statistical comparison of creep and shrinkage prediction models. *ACI Materials Journal*, 105, 610–621. <https://doi.org/10.14359/20203>.
- Bazant, Z.P., and Jirásek, M. (2018). *Creep and Hygrothermal Effects in Concrete Structures*. Solid Mechanics and Its Applications, Springer, Dordrecht. <https://doi.org/10.1007/978-94-024-1138-6>.
- Bazant, Z.P., and Wu, S.T. (1973). Dirichlet series creep function for aging concrete. *Journal of the Engineering Mechanics Division, ASCE*, 99(2), 367–387. <https://doi.org/10.1061/JMCEA3.0001741>.
- Bouras, Y., and Li, L. (2023). Utilisation of machine learning techniques to model creep behaviour of low-carbon concretes. *Buildings*, 13(9), 2252. <https://doi.org/10.3390/buildings13092252>.
- CEN (2004). *Eurocode 2: Design of Concrete Structures—Part 1-1: General Rules and Rules for Buildings* (EN 1992-1-1:2004). European Committee for Standardization, Brussels.
- Chen, T., and Guestrin, C. (2016). XGBoost: a scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. <https://doi.org/10.1145/2939672.2939785>.
- Choi, J.-S., Hong, S.-H., Yoo, S.-J., Yang, J.-M., and Yoon, Y.-S. (2025). Effect of internal curing on shrinkage and creep of self-compacting lightweight concrete under different curing conditions. *Construction and Building Materials*, 470, 140612. <https://doi.org/10.1016/j.conbuildmat.2025.140612>.
- Dokdua, W., Tangchirapat, W., and Youwai, S. (2025). Triple attention transformer architecture for time-dependent concrete creep prediction. arXiv:2506.04243. <https://doi.org/10.48550/arXiv.2506.04243>.
- Fan, Z., Zhan, Y., Xiang, T., et al. (2024). A time-series deep learning model for predicting concrete shrinkage and creep verified with in-situ and laboratory test data. *Construction and Building Materials*, 447, 138140. <https://doi.org/10.1016/j.conbuildmat.2024.138140>.
- fib (2013). *fib Model Code for Concrete Structures 2010*. Ernst & Sohn, Berlin. <https://doi.org/10.1002/9783433604090>.
- Gao, C., Luo, C., Zhang, B., Hu, Z., Tang, J., Wang, L., and Liu, J. (2025). Creep behavior evaluation of recycled aggregate concrete using machine learning technology. *Journal of Building Engineering*, 99, 111538. <https://doi.org/10.1016/j.jobbe.2024.111538>.
- Gardner, N.J., and Lockman, M.J. (2001). Design provisions for drying shrinkage and creep of normal-strength concrete. *ACI Materials Journal*, 98, 159–167. <https://doi.org/10.14359/10199>.

- Hoffman, M.D., and Gelman, A. (2014). The No-U-Turn Sampler: adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15, 1593–1623. arXiv:1111.4246. <https://doi.org/10.48550/arXiv.1111.4246>.
- Hollmann, N., Müller, S., Purucker, L., et al. (2025). Accurate predictions on small data with a tabular foundation model. *Nature*, 637, 319–326. <https://doi.org/10.1038/s41586-024-08328-6>.
- Hong, S.-H., Choi, J.-S., Yoo, S.-J., Yoo, D.-Y., and Yoon, Y.-S. (2024). Reinforcing effect of CNT on the microstructure and creep properties of high-strength lightweight concrete. *Construction and Building Materials*, 428, 136294. <https://doi.org/10.1016/j.conbuildmat.2024.136294>.
- Hoo, S.B., Müller, S., Salinas, D., and Hutter, F. (2026). From tables to time: extending TabPFN-v2 to time series forecasting. arXiv:2501.02945, version 4. <https://doi.org/10.48550/arXiv.2501.02945>.
- Hubler, M.H., Wendner, R., and Bažant, Z.P. (2015). Comprehensive database for concrete creep and shrinkage: analysis and recommendations for testing and recording. *ACI Materials Journal*, 112, 547–558. <https://doi.org/10.14359/51687453>.
- Liang, M., Chang, Z., Wan, Z., et al. (2022). Interpretable ensemble-machine-learning models for predicting creep behavior of concrete. *Cement and Concrete Composites*, 125, 104295. <https://doi.org/10.1016/j.cemconcomp.2021.104295>.
- Li, C., Lim, C.H., Wang, Q., Kong, S.Y., and Wang, X. (2025). A parameterless approach for long-term creep prediction in concrete using hybrid CNN–transformer model. *Structural Health Monitoring*, online first. <https://doi.org/10.1177/14759217251357971>.
- Lundberg, S.M., and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30. arXiv:1705.07874. <https://doi.org/10.48550/arXiv.1705.07874>.
- Müller, S., Hollmann, N., Pineda Arango, S., Grabocka, J., and Hutter, F. (2022). Transformers can do Bayesian inference. *International Conference on Learning Representations*. arXiv:2112.10510. <https://doi.org/10.48550/arXiv.2112.10510>.
- Phan, D., Pradhan, N., and Jankowiak, M. (2019). Composable effects for flexible and accelerated probabilistic programming in NumPyro. arXiv:1912.11554. <https://doi.org/10.48550/arXiv.1912.11554>.
- Potapczynski, A., Selvam, R.K., Konstantinova, T., et al. (2026). Time-aware prior fitted networks for zero-shot forecasting with exogenous variables. *Transactions on Machine Learning Research*. arXiv:2603.15802. <https://doi.org/10.48550/arXiv.2603.15802>.
- Qu, J., Holzmüller, D., Varoquaux, G., and Le Morvan, M. (2025). TabICL: A tabular foundation model for in-context learning on large data. *Proceedings of the 42nd International Conference on Machine Learning*, PMLR 267, 50817–50847. arXiv:2502.05564. <https://doi.org/10.48550/arXiv.2502.05564>.
- RILEM Technical Committee TC-242-MDC (2015). RILEM draft recommendation: TC-242-MDC multi-decade creep and shrinkage of concrete: material model and structural analysis. Model B4 for creep, drying shrinkage and autogenous shrinkage of normal and high-strength concretes with multi-decade applicability. *Materials and Structures*, 48, 753–770. <https://doi.org/10.1617/s11527-014-0485-2>.
- Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., et al. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8), 913–929. <https://doi.org/10.1111/ecog.02881>.
- Rossi, P., Tailhan, J.-L., and Le Maou, F. (2013). Creep strain versus residual strain of a concrete loaded under various levels of compressive stress. *Cement and Concrete Research*, 51, 32–37. <https://doi.org/10.1016/j.cemconres.2013.04.005>.
- Shorten, C., and Khoshgoftaar, T.M. (2019). A survey on image data augmentation for deep learning. *Journal of Big Data*, 6, 60. <https://doi.org/10.1186/s40537-019-0197-0>.
- Sousa, H., Santos, L.O., and Chryssanthopoulos, M. (2019). Quantifying monitoring requirements for predicting creep deformations through Bayesian updating methods. *Structural Safety*, 76, 40–50. <https://doi.org/10.1016/j.strusafe.2018.06.002>.

- Tao, Z., Peng, J., Liao, S., and Yao, Y. (2026). Hybrid PINN-GPR method for probabilistic prediction of concrete creep behavior. *Journal of Materials in Civil Engineering*, 38(11), 04026397. <https://doi.org/10.1061/JMCEE7.MTENG-23413>.
- Tobin, J., Fong, R., Ray, A., Schneider, J., Zaremba, W., and Abbeel, P. (2017). Domain randomization for transferring deep neural networks from simulation to the real world. *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 23–30. <https://doi.org/10.1109/IROS.2017.8202133>.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 5998–6008. arXiv:1706.03762. <https://doi.org/10.48550/arXiv.1706.03762>.
- Wang, X., Xue, Y., Xue, R., Wang, Z.-R., Wu, D., Ye, L.-X., Zhang, Y.-C., and Tian, Z.-F. (2026). Prediction of creep compliance in high strength concrete using a Feature Tokenizer Transformer with grouped validation. *Discover Applied Sciences*. <https://doi.org/10.1007/s42452-026-09280-y>.
- Wendner, R., Hubler, M.H., and Bažant, Z.P. (2015). Optimization method, choice of form and uncertainty quantification of Model B4 using laboratory and multi-decade bridge databases. *Materials and Structures*, 48, 771–796. <https://doi.org/10.1617/s11527-014-0515-0>.
- Zhang, Y., and Ling, C. (2018). A strategy to apply machine learning to small datasets in materials science. *npj Computational Materials*, 4, 25. <https://doi.org/10.1038/s41524-018-0081-z>.
- Zhao, Q., Yu, J., Geng, G., Jiang, J., and Liu, X. (2016). Effect of fiber types on creep behavior of concrete. *Construction and Building Materials*, 105, 416–422. <https://doi.org/10.1016/j.conbuildmat.2015.12.149>.